Laws of Learning: LLN, Concentration, Uniform Convergence, PAC

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Why does minimizing error on a *sample* say anything about error on the *distribution*? This page collects the results that answer that, in the order they strengthen each other: averages converge (LLN), at a quantified exponential rate (Hoeffding, Azuma), uniformly over classes (Glivenko–Cantelli), which is exactly what PAC learnability needs. `learning_theory.py` demonstrates the first three empirically.

1 Laws of large numbers

Theorem 1 (Weak LLN)**.** *If $x_1, \dots, x_n$ are i.i.d. with mean μ and finite variance, then for every $\varepsilon > 0$, $\Pr\big(|\bar{x}_n - \mu| > \varepsilon\big) \to 0$.*

Theorem 2 (Strong LLN)**.** *If the x_i are i.i.d. with $\mathbb{E}|x_1| < \infty$, then $\bar{x}_n \to \mu$* almost surely *— individual sample paths converge, not just probabilities (visible in the script's first panel).*

Remark 1 (Beyond i.i.d.)**.** *Independence can be relaxed: LLNs hold for ergodic stationary sequences (Birkhoff), for martingale-difference sequences, and under mixing conditions — what must survive is that dependence decays fast enough for averages to stabilize. Azuma's inequality below is the concentration counterpart for the martingale case, the standard tool once samples are dependent.*

2 Concentration: how fast, exactly

Theorem 3 (Hoeffding's inequality)**.** *Let $x_1, \dots, x_n$ be independent with $x_i \in [a_i, b_i]$. Then for all $\varepsilon > 0$,*

$$\Pr\Big(\big|\bar{x}_n - \mathbb{E}\bar{x}_n\big| \geq \varepsilon \Big) \;\leq\; 2\exp\Big(-\frac{2n^2\varepsilon^2}{\sum_i (b_i - a_i)^2}\Big) \;\stackrel{[0,1]}{=}\; 2e^{-2n\varepsilon^2}. \tag{1}$$

The LLN says the tail vanishes; Hoeffding says *exponentially fast*, with explicit constants and no variance knowledge — only boundedness. Inverting (1): with probability $1-\delta$, $|\bar{x}_n - \mu| \leq \sqrt{\log(2/\delta)/2n}$ — the ubiquitous $1/\sqrt{n}$ of statistics. (Proof route: Chernoff's method, $\Pr(S \geq \varepsilon) \leq e^{-\lambda\varepsilon}\mathbb{E}e^{\lambda S}$, plus Hoeffding's lemma $\mathbb{E}e^{\lambda(x - \mathbb{E}x)} \leq e^{\lambda^2(b-a)^2/8}$, then optimize λ.)

Theorem 4 (Azuma–Hoeffding)**.** *Let $M_0, M_1, \dots$ be a martingale with bounded differences $|M_k - M_{k-1}| \leq c_k$. Then*

$$\Pr\big(|M_n - M_0| \geq \varepsilon\big) \;\leq\; 2\exp\Big(-\frac{\varepsilon^2}{2\sum_k c_k^2}\Big). \tag{2}$$

Hoeffding for dependent-but-martingale increments. Its workhorse corollary (McDiarmid): any function of independent variables that changes by at most c_i when one coordinate changes concentrates around its mean — the tool behind generalization bounds for quantities that are not plain averages.

3 Uniform convergence: Glivenko–Cantelli

Hoeffding controls *one* fixed quantity. Learning selects a hypothesis from a whole class *after* seeing data, so we need convergence *simultaneously* over the class.

Theorem 5 (Glivenko–Cantelli). *For i.i.d. samples with CDF F and empirical CDF $F_n(t) = \frac{1}{n}\#\{i : x_i \le t\}$,*

$$\sup_{t\in\mathbb{R}} \big|F_n(t) - F(t)\big| \xrightarrow{a.s.} 0, \tag{3}$$

and quantitatively (Dvoretzky–Kiefer–Wolfowitz), $\Pr(\sup_t |F_n - F| > \varepsilon) \le 2e^{-2n\varepsilon^2}$ *— the same exponent as* (1)*, despite the supremum over infinitely many events.*

Definition 1 (Glivenko–Cantelli class). *A function class $\mathcal{F}$ is GC for a distribution D if $\sup_{f\in\mathcal{F}}\big|\frac{1}{n}\sum_i f(x_i) - \mathbb{E}_D f\big| \to 0$ (in probability / a.s.).*

The classical theorem says the indicators $\{\,\mathbf{1}[x \le t] : t \in \mathbb{R}\}$ form a GC class. Which classes are GC is *the* question of learning theory; the answer is measured by combinatorial capacity — finite VC dimension implies GC with rate $\sqrt{(\mathrm{VC}\log n)/n}$ — and capacity is what sample-complexity bounds charge for.

4 PAC learning

Definition 2 (PAC learnability). *A hypothesis class $\mathcal{H}$ is PAC learnable if there is an algorithm A and sample complexity $n(\varepsilon, \delta)$ such that for every distribution D and every $\varepsilon, \delta \in (0, 1)$: given $n \ge n(\varepsilon, \delta)$ i.i.d. samples, with probability at least $1 - \delta$ the output $\hat{h} = A(S)$ satisfies*

$$L_D(\hat{h}) \;\le\; \min_{h\in\mathcal{H}} L_D(h) + \varepsilon. \tag{4}$$

"Probably $(1 - \delta)$ Approximately (ε) Correct." The pieces above assemble into the fundamental result: if $\mathcal{H}$ is a GC class, the *empirical risk minimizer* $\hat{h} = \arg\min_h \frac{1}{n}\sum_i \ell(h, x_i)$ PAC-learns $\mathcal{H}$, because uniform convergence makes empirical risks a trustworthy proxy for true risks across the whole class at once. Two standard sample complexities (0–1 loss):

$$n \;=\; O\Big(\tfrac{\log|\mathcal{H}| + \log(1/\delta)}{\varepsilon^2}\Big) \text{ (finite class)}, \qquad n \;=\; O\Big(\tfrac{\mathrm{VC}(\mathcal{H}) + \log(1/\delta)}{\varepsilon^2}\Big), \tag{5}$$

with $\varepsilon^2 \to \varepsilon$ in the realizable (zero-error) case. The fundamental theorem of PAC learning closes the loop: for binary classification, *learnable* $\iff$ *finite VC dimension* $\iff$ *uniform convergence* — capacity, not cleverness of the algorithm, decides learnability.